

14/09/26

Revised Simplex Method

Standard form-I :

Recall : $\text{Max } Z = x_1 + 2x_2$
Subject $\left. \begin{aligned} x_1 + x_2 &\leq 3 \\ x_1 + 2x_2 &\leq 5 \\ 3x_1 + x_2 &\leq 6 \\ x_1, x_2 &\geq 0 \end{aligned} \right\}$

Step-I :

①

$$b = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix} = \begin{bmatrix} 3 \\ 5 \\ 6 \end{bmatrix} \text{ All } b_i \geq 0 ??$$

②

Objective fn is Max or not ??

③

Inequalities to equalities

④

we can objective fn as 1st constraint

$$Z = x_1 + 2x_2$$

$$\Rightarrow Z - x_1 - 2x_2 = 0$$

————— ①

Step-II

Convert to Matrix Representation.

$$\begin{bmatrix} \quad \quad \quad \\ \quad \quad \quad \\ \quad \quad \quad \\ A \end{bmatrix} \begin{bmatrix} \quad \quad \quad \\ \quad \quad \quad \\ \quad \quad \quad \\ X \end{bmatrix} = \begin{bmatrix} \quad \quad \quad \\ \quad \quad \quad \\ \quad \quad \quad \\ b \end{bmatrix}$$

$$\begin{array}{cccccc}
 a_0 & a_1 & a_2 & a_3 & a_4 & a_5 \\
 \left[\begin{array}{|c|} \hline \\ \hline \end{array} \right] & \left[\begin{array}{|c|} \hline \\ \hline \end{array} \right] & \left[\begin{array}{|c|} \hline \\ \hline \end{array} \right] & \left[\begin{array}{|c|} \hline \\ \hline \end{array} \right] & \left[\begin{array}{|c|} \hline \\ \hline \end{array} \right] & \left[\begin{array}{|c|} \hline \\ \hline \end{array} \right] \\
 \underbrace{\hspace{1.5cm}} & \underbrace{\hspace{1.5cm}} & \underbrace{\hspace{1.5cm}} & \underbrace{\hspace{1.5cm}} & \underbrace{\hspace{1.5cm}} & \underbrace{\hspace{1.5cm}} \\
 & \text{non-basis} & & & & \\
 & \text{variables} & & & &
 \end{array}$$

Initial Basis

$$B = (a_0, a_3, a_4, a_5)$$

$$= I_4$$

$$B^{-1} = I_4$$

Step-II check the optimality condition

$$\Delta = [\Delta_1, \Delta_2, \Delta_3, \Delta_4]$$

= first row of $B^{-1} \times$ (Additional table)

Additional Table =

$$\begin{bmatrix} a_1 & a_2 \\ | & | \end{bmatrix}$$

$\Delta_j \geq 0$ $\Rightarrow$ optimality arises.

If $\Delta_j \leq 0 \Rightarrow$ we need to find entering vector and leaving vector.

$$\Delta_k = \min (\Delta_j) \quad \forall j$$

$x_k \leftarrow$ entering variable.

Suppose a_2 is entering

Leaving variable: $\min(x_B/x_K) = x_r$.

x_r is leaving variable

[Suppose $a_4 = x_4$ is leaving]

a_0	a_3	a_4	a_5	a_2

R

check again optimality condition
 $\Delta_j \geq 0$ or not?

Standard form-II

$$\text{Min } Z = 4x_1 + 3x_2$$

Subject

$$\begin{aligned}x_1 + 2x_2 &\geq 8 \\3x_1 + 2x_2 &\geq 12 \\x_1, x_2 &\geq 0\end{aligned}$$

$\rightarrow \text{Max } Z' = -4x_1 - 3x_2$ where $Z' = (-Z)$

$$B^* = \begin{bmatrix} 1 & -c_B \\ 0 & B \end{bmatrix}$$

$$\text{Min } Z = 4x_1 + 3x_2 + 0 \cdot x_3 + 0 \cdot x_4 - M \cdot x_5 - M \cdot x_6$$

$$\text{subject to } x_1 + 2x_2 - x_3 + x_5 = 8$$

$$3x_1 + 2x_2 - x_4 + x_6 = 12$$

$x_3, x_4 \Rightarrow$ surplus variables

$x_5, x_6 \Rightarrow$ Artificial variables

$$Z - 4x_1 - 3x_2 - 0 \cdot x_3 - 0 \cdot x_4 + M \cdot x_5 + M \cdot x_6 \geq 0$$

{

$$c_B = (-M, M)$$

[standard form - I
($c_B = 0$)]

$$\frac{B^{*-1}}{\checkmark} = \begin{bmatrix} 1 & c_B B^{-1} \\ 0 & B^{-1} \end{bmatrix}$$

not identity matrix of $m+1$

↑ order $m+1$